‘projectile’ would be slowed up in its flight against the nucleus, and in many cases—especially those involving heavy nuclei—it would either come to rest at some distance from the nucleus or would be deflected so far from its path that it would never strike the nucleus. Therefore, in order actually to hit a heavier nucleus, the charged particle would have to be accelerated by means of exceedingly high voltages, which can be produced only in apparatus especially designed for this purpose. For this reason, nuclear transmutation by means of charged particles is possible, as a rule, in relatively light atoms only.

On the other hand, there is no potential barrier for uncharged particles, and therefore they can be used to transmute atoms of any mass number. Transmutations of heavier atomic nuclei by photons, i.e. gamma rays, have been observed by Bothe and his collaborators, while the transmutation by neutrons was first demonstrated by Fermi. The neutron will become incorporated in the nucleus, in many cases at any rate, and the surplus energy then will be carried off by one or more gamma-ray photons.

While this reaction signifies a change in the nucleus, it does not involve any change in the chemical properties of the atom. The chemical properties change only when the resulting nucleus is unstable because it has too many neutrons within it. In that case, a subsequent process takes place, in which a neutron changes into a proton, by the emission of an electron, with the result that the nucleus becomes a nucleus of the element whose atomic number is one higher.

Since a neutron can approach a nucleus unimpeded, the velocity plays no essential role in the process, unlike the case of charged particles. On the contrary, slow neutrons are frequently more effective than fast ones, for they remain longer in the vicinity of the nucleus, and so the probability of their eventual capture by the nuclear forces is greater than for fast neutrons.

Experiments have shown that this probability can be extraordinarily great for neutrons having a certain, but not too high, energy. According to Bethe, this becomes quite logical when we consider the neutron under the wave aspect, i.e. as a wave incident on the nucleus. The nucleus is a structure capable of vibration, and as such, it can enter into resonance with any wave it encounters, if the frequency of that wave coincides with one of its own fundamental frequencies. In this case, an exceedingly strong selective absorption of the wave occurs, a process quite familiar to us in connection with the absorption of light. Now the frequency of the wave is a function of the velocity of the neutron, and therefore there exists a very definite velocity where the requirements for resonance are fulfilled and the wave is especially strongly absorbed by the nucleus. But translated back again into terms of the particle aspect, this means that the probability of the capture of the neutron by the nucleus is a very high one. This dependence on the velocity is often